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Experiment 1 · Create Static Home Web Page

Mapped Course Outcome: CO1 • Scheduled: Week 1

Aim

Write a html code to make a static home web page containing three frames.

Theory

A static homepage can be divided into a top navigation section, a left catalogue section, and a right description section. The submitted program uses CSS Flexbox to create this three-section layout in one HTML document.

HTML Code

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Homepage Design</title>
  <style>
    body { margin: 0; font-family: Arial; }
    #top { text-align: center; padding: 10px; background-color: rgb(94, 136, 153);
height: 20vh; }
    #container { display: flex; height: 80vh; }
    #catalogue { width: 25%; background-color: rgb(92, 117, 125); padding: 20px; }
    #description { width: 75%; background-color: lightgray; padding: 20px; }
    a { display: block; margin: 15px 0; text-decoration: none; color: blue; }
  </style>
</head>
<body>
  <div id="top">
    <h1>GL Bajaj Institute of Technology and Management</h1>
    <a href="#" style="display: inline">Home</a>
    <a href="#" style="display: inline">About</a>
    <a href="#" style="display: inline">Contact</a>
  </div>

  <div id="container">
    <div id="catalogue">
      <h2>Catalogue Links</h2>
      <a href="#cse">CSE </a>
      <a href="#ece">ECE</a>
      <a href="#eee">EEE</a>
      <a href="#civil">CIVIL</a>
    </div>

    <div id="description">
      <h2>Welcome</h2>
      <p>Select a branch from the catalogue to view the details.</p>

      <h3 id="cse">CSE</h3>
      <p>
        Computer Science and Engineering (CSE) focuses on the design,
        development, and management of software and hardware systems.
      </p>

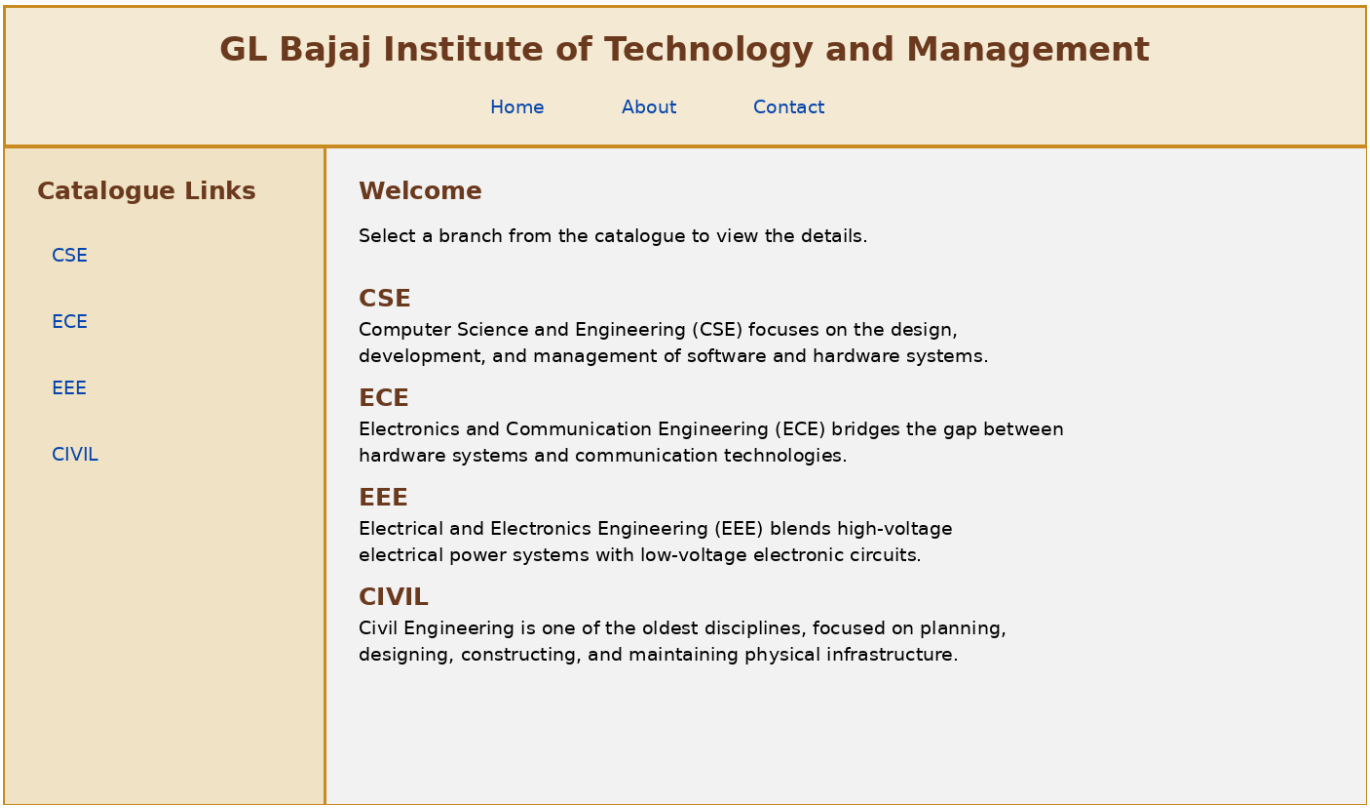
      <h3 id="ece">ECE</h3>
      <p>
        Electronics and Communication Engineering (ECE) bridges the gap
        between hardware systems and communication technologies.
      </p>

      <h3 id="eee">EEE</h3>
      <p>
        Electrical and Electronics Engineering (EEE) blends high-voltage
        electrical power systems with low-voltage electronic circuits.
      </p>

      <h3 id="civil">CIVIL</h3>
      <p>
        Civil Engineering is one of the oldest disciplines, focused on
        planning, designing, constructing, and maintaining physical
        infrastructure.
      </p>
    </div>
  </div>
</body>
</html>
```

Experiment 1 · Sample Output

Sample Input / Output



Applications

- Admin Dashboards: Displaying persistent navigation and a central workspace.
- College Portals: Presenting department or branch information through catalogue links.
- E-Commerce Catalogs: Organizing categories and related information in a structured layout.

Experiment 2 · Login Page Design

Mapped Course Outcome: CO1 · Scheduled: Week 2

Aim

Write a HTML code to design a login page with Username, Password fields, Submit and Reset buttons, plus navigation links.

Theory

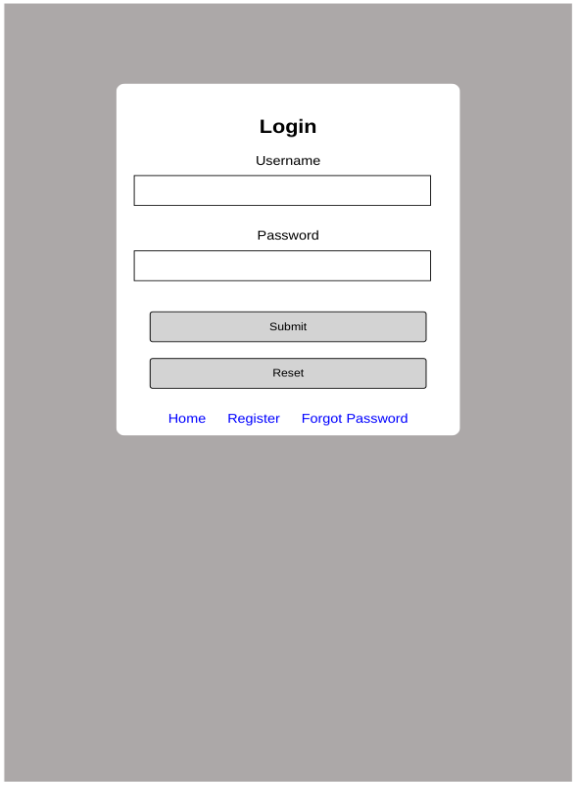
A login page provides a simple interface for user authentication. It contains input fields for username and password, action buttons for submitting or resetting the form, and navigation links for moving to related pages such as Home, Register, and Forgot Password.

HTML Code

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Login Page</title>
    <style>
        body{
            text-align: center;
            font-family: Arial, sans-serif;
            background-color: rgb(173, 168, 168);
        }
        #login-page{
            width: 350px;
            height:400px;
            margin:100px auto;
            padding: 20px;
            background-color: white;
            border-radius: 10px;
        }
        input{
            width: 90%;
            padding:10px;
            margin: 10px 0;
        }
        button{
            padding: 10px 20px;
            margin: 10px 5px;
            cursor: pointer;
        }
        a{
            margin: 10px;
            text-decoration: none;
            color: blue;
        }
    </style>
  </head>
  <body>
    <div id="login-page">
      <h2>Login</h2>
      <form>
        <label for="username">Username </label>
        <input type="text" id="username" placeholder="Enter Username " required /><br />
        <label for="password">Password </label>
        <input type="password" id="password" placeholder="Enter your Password" required /><br />
        <button type="submit">Submit</button>
        <button type="reset">Reset</button>
      </form>
      <br />
      <a href="#">Home</a>
      <a href="#">Register</a>
      <a href="#">Forgot Password</a>
    </div>
  </body>
</html>
```

Experiment 2 · Sample Output

Sample Input / Output



Applications

• User Authentication: Providing a basic interface for entering login credentials.
• Website Navigation: Linking users to Home, Register, and Forgot Password pages.
• Secure Access Interfaces: Serving as the front end of a user sign-in workflow.

Experiment 3 · Catalogue Page Display

Mapped Course Outcome: CO1 · Scheduled: Week 3

Aim

Write a HTML code to design a catalogue page that displays books in a table with Cover Snapshot, Author, Publisher, Price, and Add-to-Cart button.

Theory

A catalogue page presents available books in a structured tabular format. Each book entry contains a cover snapshot, book details such as author and publisher, the price, and an Add-to-Cart button for selecting the book.

HTML Code

```
<!doctype html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Catalogue Page</title>
<style>
body { font-family: Arial, sans-serif; margin: 30px; background-color: #f5f5f5; }
h1 { text-align: center; color: #333; }
table { width: 100%; border-collapse: collapse; background-color: white; }
th, td { border: 1px solid #ccc; padding: 12px; text-align: center; }
th { background-color: #333; color: white; }
img { width: 80px; height: 110px; object-fit: cover; }
.price { font-weight: bold; color: green; }
button { background-color: #007bff; color: white; border: none; padding: 8px 15px;
cursor: pointer; border-radius: 4px; }
button:hover { background-color: #0056b3; }
</style>
</head>
<body>
<h1>Book Catalogue</h1>
<table>
<thead>
<tr>
<th>Cover Snapshot</th><th>Book Name</th><th>Author</th><th>Publisher</th><th>Price</th>
<th>Add to Cart</th></tr>
</thead>
<tbody>
<tr>
<td><img src="https://placehold.co/80x110?text=Book+1" alt="Book 1" /></td>
<td>Web Technologies: Black Book</td>
<td>Kogent Learning Solutions Inc.</td>
<td>Dreamtech Press</td>
<td class="price">■650</td>
<td><button>Add to Cart</button></td>
</tr>
<tr>
<td><img src="https://placehold.co/80x110?text=Book+2" alt="Book 2" /></td>
<td>Programming the World Wide Web</td>
<td>Robert W. Sebesta</td>
<td>Pearson Education</td>
<td class="price">■600</td>
<td><button>Add to Cart</button></td>
</tr>
<tr>
<td><img src="https://placehold.co/80x110?text=Book+3" alt="Book 3" /></td>
<td>Web Technologies</td>
<td>Uttam K. Roy</td>
<td>Oxford University Press</td>
<td class="price">■550</td>
<td><button>Add to Cart</button></td>
</tr>
<tr>
<td><img src="https://placehold.co/80x110?text=Book+4" alt="Book 4" /></td>
<td>HTML &amp;amp; CSS: The Complete Reference</td>
<td>Thomas A. Powell</td>
<td>McGraw Hill Education</td>
<td class="price">■700</td>
<td><button>Add to Cart</button></td>
</tr>
</tbody>
</table>
</body>
</html>
```

Experiment 3 · Sample Output

Sample Input / Output

Cover Snapshot	Book Name	Author	Publisher	Price	Add-to-Cart
	Web Technologies: Black Book	Kogent Learning Solutions Inc.	Dreamtech Press	₹650	Add to Cart
	Programming the World Wide Web	Robert W. Sebesta	Pearson Education	₹600	Add to Cart
	Web Technologies	Uttam K. Roy	Oxford University Press	₹550	Add to Cart
	HTML & CSS: The Complete Reference	Thomas A. Powell	McGraw Hill Education	₹700	Add to Cart

Applications

- Online Bookstores: Displaying available books with prices and purchase controls.
- Catalogue Management: Presenting book information in a clear, structured table.
- Shopping Interfaces: Allowing users to select books through Add-to-Cart buttons.

Experiment 4 · Cart Page

Mapped Course Outcome: CO1 · Scheduled: Week 4

Aim

Write a HTML code to design a cart page that shows details of books added to the cart, including book information, price, quantity, total amount, remove action, and checkout option.

Theory

A cart page displays the books selected by a user before checkout. It presents the book details, author, publisher, price, quantity, item total, and an action to remove a book. A grand total and Proceed to Checkout button complete the cart interface.

HTML Code

```
<!doctype html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Cart Page</title>
<style>
body { font-family: Arial, sans-serif; margin: 30px; background-color: #f5f5f5; }
h1 { text-align: center; }
table { width: 100%; border-collapse: collapse; background-color: white; }
th, td { border: 1px solid #ccc; padding: 12px; text-align: center; }
th { background-color: #333; color: white; }
img { width: 60px; height: 80px; object-fit: cover; }
.price { color: green; font-weight: bold; }
.total { text-align: right; font-size: 20px; font-weight: bold; margin-top: 20px; }
button { background-color: red; color: white; border: none; padding: 8px 15px;
border-radius: 4px; cursor: pointer; }
button:hover { background-color: darkred; }
.checkout { background-color: green; padding: 10px 20px; font-size: 16px; }
.checkout:hover { background-color: darkgreen; }
</style>
</head>
<body>
<h1>Shopping Cart</h1>
<table>
<thead>
<tr>
<th>Book</th>
<th>Author</th>
<th>Publisher</th>
<th>Price</th>
<th>Quantity</th>
<th>Total</th>
<th>Action</th>
</tr>
</thead>
<tbody>
<tr>
<td><img src="https://placehold.co/60x80?text=Book+1" alt="Book 1" /><br />
Web Technologies: Black Book</td>
<td>Kogent Learning Solutions Inc.</td>
<td>Dreamtech Press</td>
<td class="price">■650</td>
<td>1</td>
<td class="price">■650</td>
<td><button>Remove</button></td>
</tr>
<tr>
<td><img src="https://placehold.co/60x80?text=Book+2" alt="Book 2" /><br />
Programming the World Wide Web</td>
<td>Robert W. Sebesta</td>
<td>Pearson Education</td>
<td class="price">■600</td>
<td>1</td>
<td class="price">■600</td>
<td><button>Remove</button></td>
</tr>
<tr>
<td><img src="https://placehold.co/60x80?text=Book+3" alt="Book 3" /><br />
Web Technologies</td>
<td>Uttam K. Roy</td>
<td>Oxford University Press</td>
<td class="price">■550</td>
<td>1</td>
<td class="price">■550</td>
<td><button>Remove</button></td>
</tr>
<tr>
<td><img src="https://placehold.co/60x80?text=Book+4" alt="Book 4" /><br />
HTML &amp;amp; CSS: The Complete Reference</td>
<td>Thomas A. Powell</td>
<td>McGraw Hill Education</td>
<td class="price">■750</td>
<td>1</td>
<td class="price">■750</td>
<td><button>Remove</button></td>
</tr>
</tbody>
</table>
<div class="total">Grand Total: ■2550</div>
<div style="text-align: right; margin-top: 15px">
<button class="checkout">Proceed to Checkout</button>
</div>
</body>
</html>
```

Experiment 4 · Sample Output

Sample Input / Output

Book	Author	Publisher	Price	Quantity	Total	Action
 Web Technologies: Black Book	Kogent Learning Solutions Inc.	Dreamtech Press	■650	1	■650	Remove
 Programming the World Wide Web	Robert W. Sebesta	Pearson Education	■600	1	■600	Remove
 Web Technologies	Uttam K. Roy	Oxford University Press	■550	1	■550	Remove
 HTML & CSS: The Complete Reference	Thomas A. Powell	McGraw Hill Education	■750	1	■750	Remove

Grand Total: ■2550

Proceed to Checkout

Applications

- E-commerce websites: Showing selected products before purchase.
- Online bookstores: Managing selected books, quantities, and item totals.
- Checkout interfaces: Providing a clear final total and checkout action.

Experiment 5 · Registration Page

Mapped Course Outcome: CO1 · Scheduled: Week 5

Aim

Create a registration form with fields: Name, Password, Email, Phone, Sex, DOB, Languages Known, and Address.

Theory

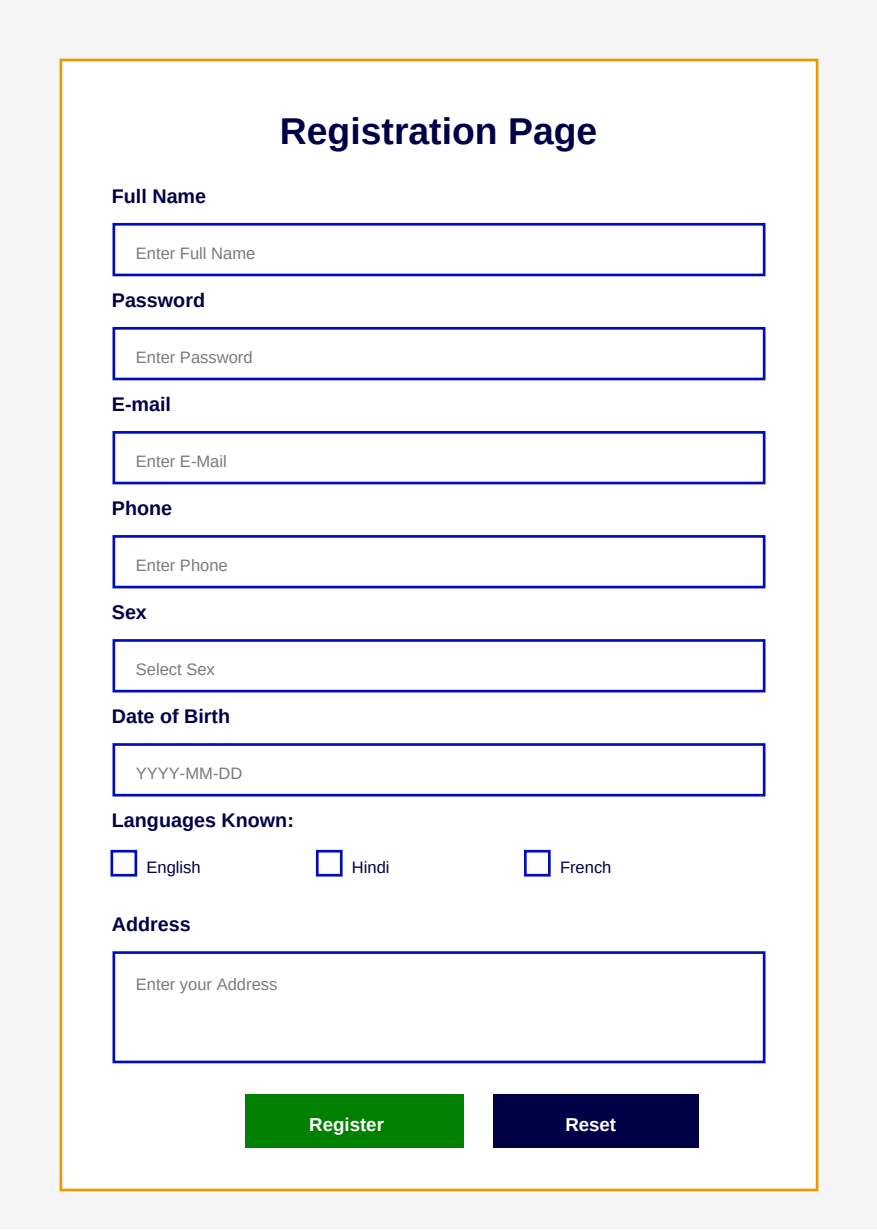
A registration page collects user information through HTML form controls. Text, password, email, phone, date, selection, checkboxes, and textarea controls are used to collect the required details. Submit and Reset buttons provide basic form actions.

HTML Code

```
<!doctype html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Registration Page</title>
<style>
body { font-family: Arial, sans-serif; background-color: #f5f5f5;
margin: 0; padding: 30px; }
h1 { text-align: center; color: #333; }
form { width: 400px; margin: auto; padding: 25px;
background-color: white; border: 1px solid #ccc; border-radius: 8px; }
label { display: block; margin-top: 12px; margin-bottom: 5px;
font-weight: bold; }
input[type="text"], input[type="password"], input[type="email"],
input[type="tel"], input[type="date"], textarea {
width: 100%; padding: 8px; box-sizing: border-box; }
select { width: 100%; padding: 8px; box-sizing: border-box; }
.language-group label { display: inline; font-weight: normal;
margin-right: 15px; }
textarea { height: 80px; resize: vertical; }
button[type="reset"] { background-color: #777; }
button { background-color: green; color: white; text-align: center;
padding: 10px 20px; margin-top: 20px; border: none;
border-radius: 4px; cursor: pointer; }
</style>
</head>
<body>
<h1>Registration Page</h1>
<form>
<label for="name">Full Name</label>
<input type="text" id="name" name="name" placeholder="Enter Full Name" required />
<label for="password">Password</label>
<input type="password" id="password" name="password" placeholder="Enter Password" required />
<label for="email">E-mail</label>
<input type="email" id="email" name="email" placeholder="Enter E-Mail" required />
<label for="phone">Phone</label>
<input type="tel" id="phone" name="phone" placeholder="Enter Phone" required />
<label for="gender">Sex</label>
<select id="gender" name="gender" required>
<option value="" disabled selected>Select Sex</option>
<option value="male">Male</option>
<option value="female">Female</option>
<option value="other">Other</option>
</select>
<label for="dob">Date of Birth</label>
<input type="date" id="dob" name="dob" required />
<label>Languages Known:</label>
<div class="language-group">
<input type="checkbox" id="english" name="language" value="English" />
<label for="english">English</label>
<input type="checkbox" id="hindi" name="language" value="Hindi" />
<label for="hindi">Hindi</label>
<input type="checkbox" id="french" name="language" value="French" />
<label for="french">French</label>
</div>
<label for="address">Address</label>
<textarea id="address" name="address" placeholder="Enter your Address" required></textarea>
<button type="submit">Register</button>
<button type="reset">Reset</button>
</form>
</body>
</html>
```

Experiment 5 · Sample Output

Registration Form



Applications

· User registration on websites and web applications.
· Collecting personal details before account creation.
· Form-based data entry using standard HTML controls.